Computational Techniques For Discounted, Finite-Space MDPs

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Computational Techniques

- Value Iteration

- Policy Iteration

Value Iteration

1. Make an initial guess V_0, set $k = 1$
2. $V_{k+1} = T(V_k)$
3. $k \leftarrow k + 1$, repeat steps 2-3 till "convergence"

- From the contraction mapping theorem, we have convergence
  - In practice, we have to use some stopping criterion
- Let V_K be the output. Our guess for the optimal policy is

$$\pi(i) \in \arg\min_a r(i,a) + \sum_j P_{ij}(a) V_K(j)$$

  - One can obtain bounds on the suboptimality of this policy as in the notes on the contraction mapping theorem

Policy Iteration

1. Start with a policy π_0. Set $k = 0$
2. Compute the value function V_{π_k} of π_k: it is the unique fixed point of the equation $V = T_{\pi_k}(V)$
   - Recall $T_\pi(V)(i) = r\big(x, \pi(i)\big) + \alpha \sum_j P_{ij}\big(\pi(i)\big) V(j)$
3. $k \leftarrow k + 1$, update the policy:

$$\pi_k(i) \in \arg\max_a r(i,a) + \alpha \sum_j P_{ij}(a)\, V(j)$$

4. Repeat 3-4 till "convergence"

Compact Notation for Policy Iteration

- The two main steps of PI can be written as
  - $V_{\pi_k} = T_{\pi_k}(V_{\pi_k})$: Find the value function of the current policy
  - $T_{\pi_{k+1}}(V_{\pi_k}) = T(V_{\pi_k})$: Find an updated policy by performing the minimization in the Bellman equation using the current estimate of the value function

- The convergence of policy iteration and related algorithms uses the following montonicity property of the T and T_π mappings:
  - Monotonicity: $T(V_1) \leq T(V_2)$ and $T_\pi(V_1) \leq T_\pi(V_2)$ if $V_1 \leq V_2$

Policy Improvement under PI

- Theorem: Under PI, the following statements hold:
  1. $V_{\pi_{k+1}} \geq V_{\pi_k}$, i.e., the policy "improves" at each step, and
  2. if $V_{\pi_{k+1}} = V_{\pi_k}$, then π_k is an optimal policy.

a) $T_{\pi_{k+1}}(V_{\pi_k}) = T(V_{\pi_k}) \geq T_{\pi_k}(V_{\pi_k}) = V_{\pi_k}$

- The first equality is the policy update step of PI. Inequality follows from the maximization in T. The final equality is from the definition of T_{π_k} and V_{π_k}

b) Apply $T_{\pi_{k+1}}$ to both sides of the inequality: using monotonicity and (a) above, $T^2_{\pi_{k+1}}(V_{\pi_k}) \geq T_{\pi_{k+1}}(V_{\pi_k}) \geq V_{\pi_k}$

c) Applying $T_{\pi_{k+1}}$ repeatedly, we get $T^N_{\pi_{k+1}}(V_{\pi_k}) \geq V_{\pi_k}$

- Taking the limit as $N \to \infty$, the LHS becomes $V_{\pi_{k+1}}$ from the contraction property of $T_{\pi_{k+1}}$, proving (1)

d) If $V_{\pi_{k+1}} = V_{\pi_k}$, then $T_{\pi_{k+1}}(V_{\pi_k}) = T(V_{\pi_k}) \Leftrightarrow T_{\pi_{k+1}}(V_{\pi_{k+1}}) = T(V_{\pi_{k+1}})$, which is equivalent to $V_{\pi_{k+1}} = T(V_{\pi_{k+1}})$, which implies optimality! Thus, (2) is proved

Policy Iteration vs Value Iteration

- The previous theorem states that either you get an improved policy at each step of PI or you have found the optimal policy
  - So, in principle, PI converges in a finite number of steps if the state and action spaces are finite
  - However, at each step, one has to compute V_{π_k} This can be done iteratively as in value iteration $V_{t+1} = T_{\pi_k}(V_t)$. Think of this as an inner loop, this inner loop may take a long time to produce an accurate value for V_{π_k}

Linear Programming

- Consider the linear program

$$\min_{V} \sum_{i} V_i$$

s.t.

$$V_i \geq r(i,a) + \alpha \sum_{j} P_{ij}(a) V_j \quad \forall i, a$$

- We claim that the solution to the above LP yields V^*

Proof

- Suppose at the optimal solution, for some i

$$V(i) > r(i,a) + \alpha \sum_j P_{ij}(a) V_j \quad \forall a$$

- Then one can decrease $V(i)$ slightly while satisfying the constraint thus decreasing the objective

- Thus, at the optimal solution $V(i) \geq r(i,a) + \alpha \sum_j P_{ij}(a) V_j \quad \forall a$, with equality for at least one a, i.e., $V = T(V)$